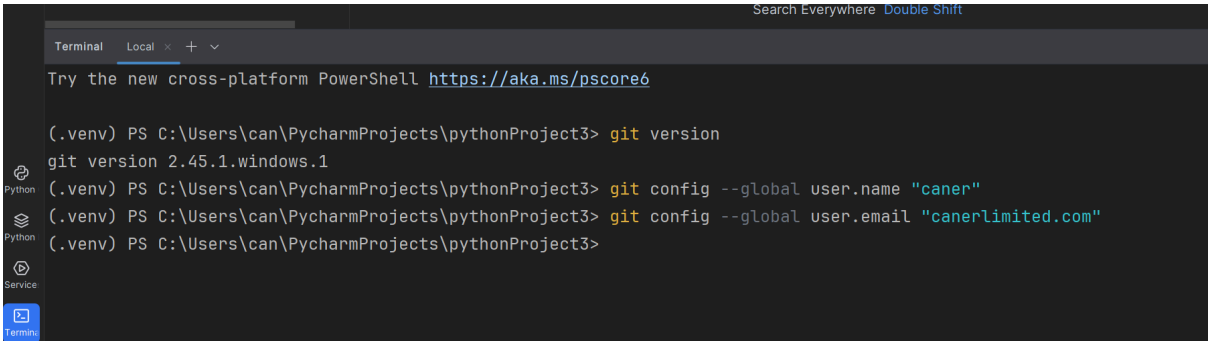
Software Build Automation Tools - GIT Automation Lab Report

TASK1

Caner akcasu
github: https://github.com/Canerakcasu/Software-build-automation-tools

We **checked if Git is installed** using `git --version`. If it wasn't installed, we installed it based on the operating system.

We **configured Git with our name and email** using `git config --global user.name` and `git config --global user.email`



Created a new folder (`mkdir git-automation-lab`).

Navigated into the folder (`cd git-automation-lab`).

Initialized a **Git repository** (`git init`), which created a `.git` folder to track changes.

Created a **README.md file** for project documentation.

Staged (`git add`) and committed (`git commit -m`) the file.

```
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3> cd .\git-automation-lab\
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git init
Initialized empty Git repository in C:/Users/can/PycharmProjects/pythonProject3/git-automation-lab/.git/
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab>  echo "# Git Automation Lab" > README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git add README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git commit -m "added readme file"
[master (root-commit) 6ec17c5] added readme file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab>
```

Created and switched to `feature-branch`.

Modified `README.md` and committed the changes.

Merged `feature-branch` into `main`.

Pushed everything to GitHub.

```
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git merge feature-branch
Already up to date.
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git remote add origin https://github.com/Canerakcasu/Software-build-automation-tools
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Canerakcasu/Software-build-automation-tools.git'
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git branch -M main
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab>
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git remote add origin https://github.com/Canerakcasu/Software-build-automation-tools
.git
error: remote origin already exists.
```

Created and switched to `feature-branch`.

Modified `README.md` and committed the changes.

Merged `feature-branch` into `main`.

Pushed everything to GitHub.

```
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 522 bytes | 522.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Canerakcasu/Software-build-automation-tools.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab>
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab>
```

Created two branches that modified the same line.

Merged `branch1` into `main` successfully.

Merging `branch2` caused a conflict, which we manually resolved.

```
Terminal   Local
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\git-automation-lab> cd ..
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
Receiving objects: 100% (6/6), done.
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3> cd .\Software-build-automation-tools\
Switched to a new branch 'branch1'
Switched to a new branch 'branch2'
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> echo "Conflicting change from branch1" >> README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git add README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git commit -m "update branch 1"
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git checkout branch2
Already on 'branch2'
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> echo "Conflicting change from branch2" >> README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git add README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git commit -m "Update from branch2"
 1 file changed, 0 insertions(+), 0 deletions(-)
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git merge branch1
pythonProject3                                                                                  tabnine Basic
```

```
 1 file changed, 0 insertions(+), 0 deletions(-)
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git add .\README.md
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git commit -m "resolved merge"
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 560 bytes | 560.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Canerakcasu/Software-build-automation-tools.git
   3b281c1..c72b530  main -> main
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools>
```

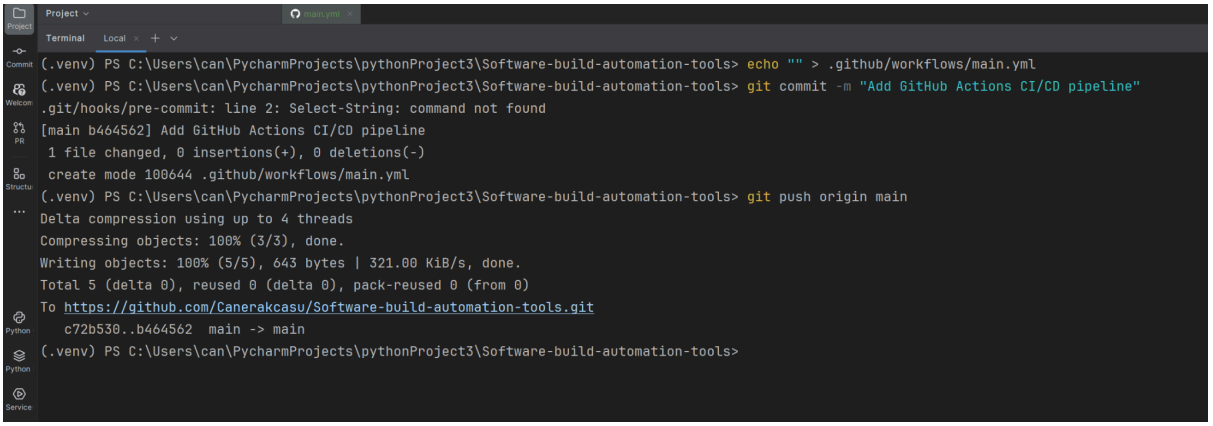
i failed on this part

```
Terminal   Local
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools\.git\hooks> touch pre-commit
touch : The term 'touch' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a
was included, verify that the path is correct and try again.
At line:1 char:1
+ touch pre-commit
+ ~~~~~
    + CategoryInfo          : ObjectNotFound: (touch:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools\.git\hooks> nano pre-commit
At line:1 char:1
+ nano pre-commit
+ ~~~~
    + CategoryInfo          : ObjectNotFound: (nano:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools\.git\hooks> New-Item pre-commit -ItemType File
```

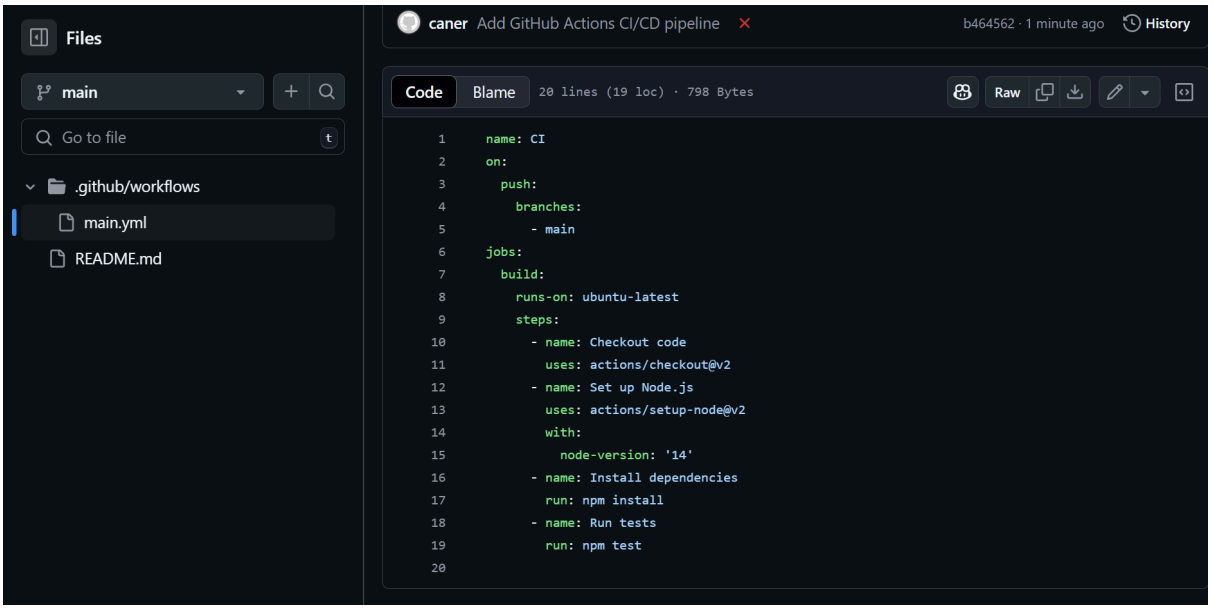
Created a GitHub Actions workflow (`main.yml`) to automate CI/CD.

Configured the pipeline to install Node.js, dependencies, and run tests on push events to the `main` branch.

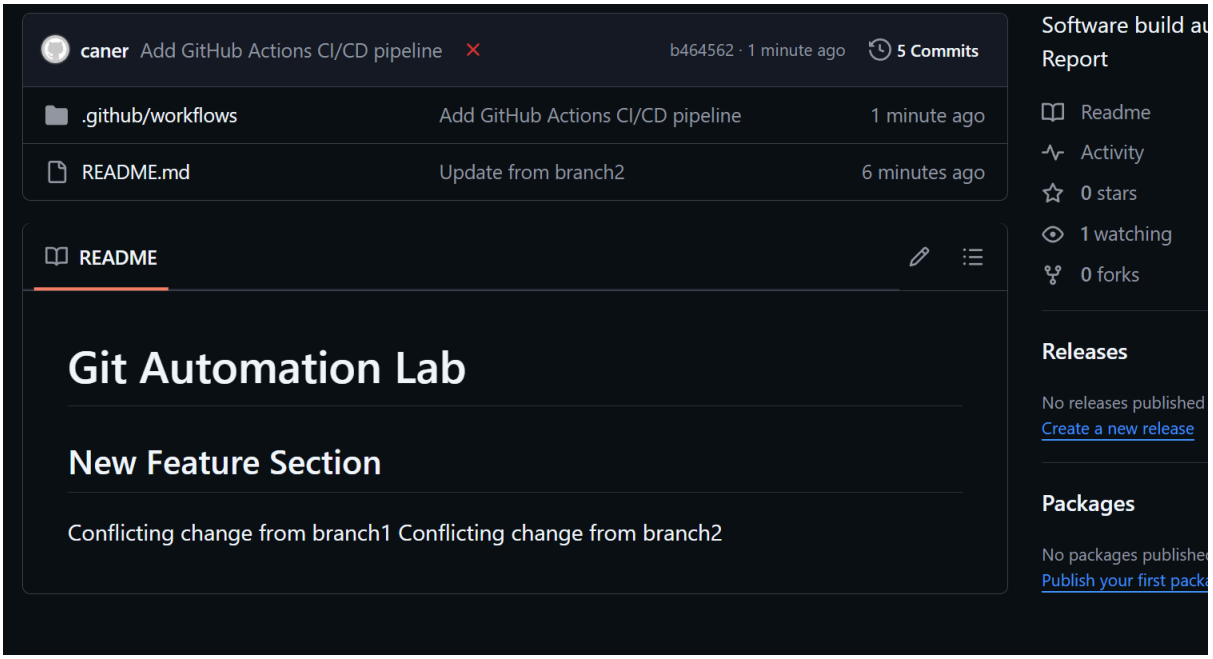
Committed and pushed the workflow to GitHub.



```
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> echo "" > .github/workflows/main.yml
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git commit -m "Add GitHub Actions CI/CD pipeline"
.git/hooks/pre-commit: line 2: Select-String: command not found
[main b464562] Add GitHub Actions CI/CD pipeline
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 .github/workflows/main.yml
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools> git push origin main
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 643 bytes | 321.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Canerakcasu/Software-build-automation-tools.git
   c72b530..b464562  main -> main
(.venv) PS C:\Users\can\PycharmProjects\pythonProject3\Software-build-automation-tools>
```



```
name: CI
on:
  push:
    branches:
      - main
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v2
      - name: Set up Node.js
        uses: actions/setup-node@v2
        with:
          node-version: '14'
      - name: Install dependencies
        run: npm install
      - name: Run tests
        run: npm test
```


caner Add GitHub Actions CI/CD pipeline
b464562 · 1 minute ago
5 Commits
.github/workflows
Add GitHub Actions CI/CD pipeline
1 minute ago
README.md
Update from branch2
6 minutes ago
README
Git Automation Lab
New Feature Section
Conflicting change from branch1 Conflicting change from branch2
Software build au
Report
Readme
Activity
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages publishe
Publish your first packa